

Smoking, BMI, and Age as Determinants of Healthcare Expenditure: An Applied Econometric Analysis

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1. Introduction

1.1 Background

Healthcare expenditure is shaped by a mix of demographic and lifestyle factors, and identifying which factors matter, and how, is useful to insurers, health economists, and policymakers. Smoking, body mass index (BMI), and age are three of the most studied individual-level drivers of medical cost, but their relationship with expenditure is rarely fixed. The association between smoking and cost, for example, may depend on a person's age or BMI.

Expenditure also varies substantially across individuals for reasons this dataset can only partly capture. Age and smoking status plausibly account for some of that variation directly, while BMI adds a further individual characteristic worth testing.

This project goes beyond a simple main-effects model by testing whether the association between smoking and healthcare expenditure changes across BMI and age.

1.2 Objectives

- Examine the association between smoking and healthcare expenditure.
- Assess the relationship between age, BMI, and healthcare expenditure.
- Examine whether the smoking–expenditure association varies with BMI and age.
- Evaluate the final OLS model using econometric diagnostic tests.
- Assess robustness using a Gamma GLM with a log link.

1.3 Research Question

How are smoking, BMI, and age associated with healthcare expenditure, and does the association between smoking and expenditure vary across age and BMI?

1.4 Interpretation

The analysis uses observational, cross-sectional data. The results describe statistical associations, not causal effects.

2. Data and Methodology

2.1 Dataset and Variables

The Medical Cost Personal Dataset records age, sex, BMI, number of children, smoking status, region, and individual healthcare expenditure, measured in USD.

Variable	Role
charges	Annual healthcare expenditure (USD)
log_charges	Natural log of expenditure; OLS dependent variable
age_c	Mean-centred age
bmi_c	Mean-centred BMI
smoker	Smoking status
sex	Sex
children	Number of children
region	Residential region
smoker × BMI	Smoking–BMI interaction
smoker × age	Smoking–age interaction

2.2 Data Cleaning

Step	Result
Initial observations	1,338
Missing values	0
Duplicate observations	1
Duplicates removed	1
Final sample	1,337

Categorical variables were converted to factors. One duplicate observation was removed and no missing values were found.

2.3 Variable Preparation

Healthcare expenditure was log-transformed because the raw distribution is right-skewed. The log-expenditure summary has a minimum of 7.023, a median of 9.147, a mean of 9.100, and a maximum of 11.063.

Age and BMI were mean-centered before the interaction terms were built, which makes the main-effect coefficients easier to interpret once interactions are added to the model.

2.4 Econometric Approach

Seven OLS models were estimated sequentially, starting with smoking alone and progressively adding age, BMI, demographic controls, and interaction terms. The final specification was:

$$\log(\text{charges}) = \beta_0 + \beta_1 \text{age_c} + \beta_2 \text{bmi_c} + \beta_3 \text{smoker} + \beta_4 \text{sex} + \beta_5 \text{children} + \beta_6 \text{region} + \beta_7 (\text{smoker} \times \text{bmi_c}) + \beta_8 (\text{smoker} \times \text{age_c}) + \varepsilon$$

3. Exploratory Data Analysis

3.1 Distribution of Healthcare Expenditure

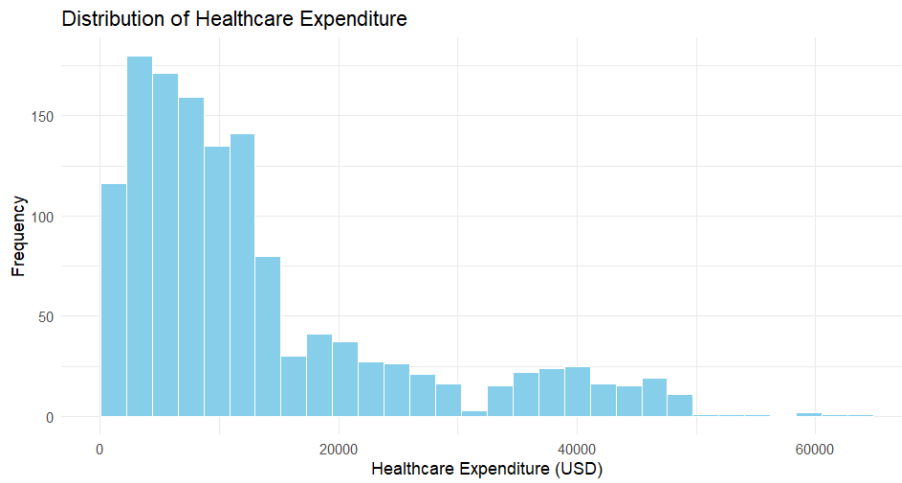


Figure 1. Distribution of Healthcare Expenditure

Healthcare expenditure is strongly right-skewed. Most individuals incur a few thousand dollars a year in charges, while a smaller group, made up largely of smokers (shown in Section 3.3), incurs far higher costs.

3.2 Distribution of Log Healthcare Expenditure

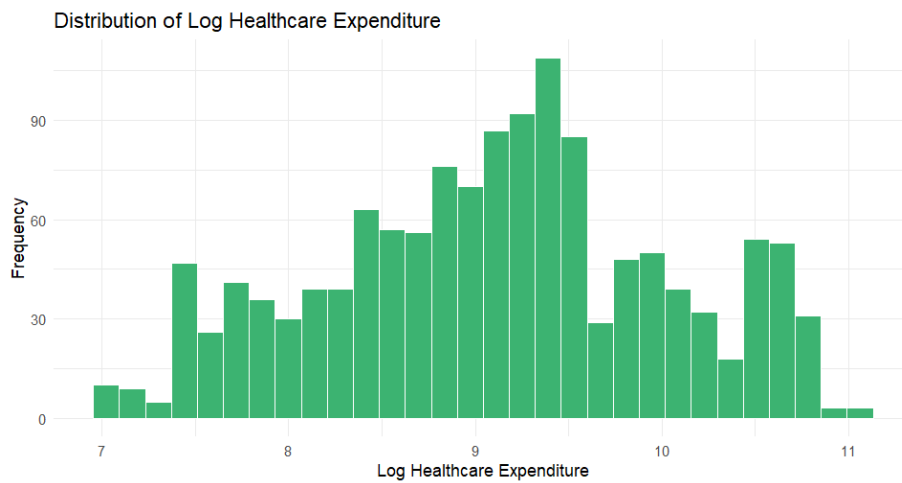


Figure 2. Distribution of Log Healthcare Expenditure

The log transformation brings the distribution much closer to symmetric, which supports using it as the dependent variable in the OLS models that follow.

3.3 Healthcare Expenditure by Smoking Status



Figure 3. Healthcare Expenditure by Smoking Status

Smokers show a substantially higher median expenditure and a wider spread than non-smokers. Smoking status alone explains a large share of the variation in cost, consistent with Model 1's R^2 of 0.4432 using smoking status as the only predictor.

3.4 Age vs Healthcare Expenditure by Smoking Status

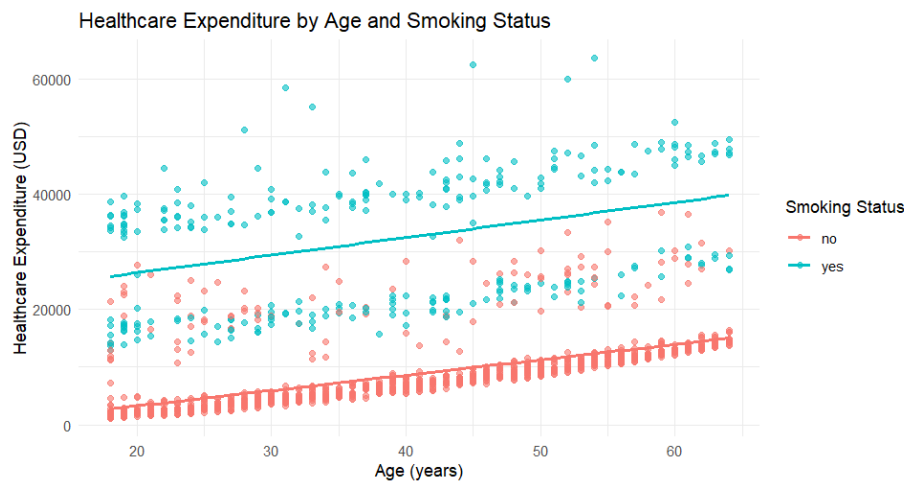


Figure 4. Healthcare Expenditure by Age and Smoking Status

Expenditure rises with age for both smokers and non-smokers, but the gap between the two groups narrows across the age range rather than staying constant. That pattern is what motivates the Smoking $\times$ Age interaction term introduced in Section 4.

4. Regression Results

4.1 Model Progression

Model	Specification	R^2	Adjusted R^2
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1	Smoking	0.4432	0.4428
2	Smoking + Age	0.7393	0.7389
3	Smoking + Age + BMI	0.7443	0.7438
4	+ Sex + Children + Region	0.7676	0.7662
5	+ Smoking $\times$ BMI	0.7832	0.7818
6	+ Smoking $\times$ Age	0.8051	0.8038
7	Both interactions	0.8246	0.8232

Fit improves steadily across the sequence, with R^2 rising from 0.4432 in Model 1 to 0.8246 in Model 7. The biggest single jump comes from adding age (Model 2), and the two interaction terms together add another 8 points of R^2 on top of the control-adjusted Model 4.

4.2 Final OLS Model

Variable	Coefficient	Std. Error	p-value
Age	0.041534	0.000850	$< 2.2e-16$
BMI	0.001177	0.002045	0.5650
Smoking: yes	1.533809	0.026351	$< 2.2e-16$
Male	-0.084992	0.021250	6.69e-05
Children	0.106094	0.008786	$< 2.2e-16$
Region: northwest	-0.063657	0.030380	0.0363
Region: southeast	-0.144961	0.030522	2.26e-06
Region: southwest	-0.150922	0.030472	8.26e-07
Smoking $\times$ BMI	0.051016	0.004207	$< 2.2e-16$
Smoking $\times$ Age	-0.033361	0.001888	$< 2.2e-16$

Final model: $R^2 = 0.8246$; adjusted $R^2 = 0.8232$; residual standard error = 0.3863; $F = 623.2$, $p < 2.2e-16$.

4.3 Interpretation of Main Variables

At the sample's average age and BMI:

- Age: each additional year is associated with roughly a 4.2% increase in expenditure ($e^{0.0415} - 1$), holding other variables fixed ($p < .001$).
- Smoking: the strongest single association in the model. At average age and BMI, smoking is associated with roughly a 364% increase in expenditure relative to non-smokers ($e^{1.5338} - 1$), before accounting for how that gap shifts with age and BMI ($p < .001$).
- BMI: not statistically significant as a standalone term ($p = 0.565$). This does not mean BMI is irrelevant to expenditure; its interaction with smoking status is highly significant, discussed in Section 4.4.
- Sex, children, region: smaller but statistically significant associations. Male sex and residence in the southern regions are associated with lower expenditure; each additional child is associated with higher expenditure.

4.4 Interaction Effects

Because the final model includes interaction terms, the smoking coefficient should not be read as a single, universal effect. It represents the estimated smoking association at the mean (centred) values of age and BMI. The Smoking $\times$ BMI term tests whether this association changes with BMI; the Smoking $\times$ Age term tests whether it changes with age.

The Smoking $\times$ BMI interaction is positive and significant (coefficient = 0.0510, $p < .001$): the estimated smoking–expenditure association strengthens as BMI rises. The Smoking $\times$ Age interaction is negative and significant (coefficient = -0.0334 , $p < .001$): the association weakens as age rises. One plausible reading is that non-smokers' costs climb steadily with age and gradually close some of the relative gap with smokers, even though smokers' absolute costs stay higher across the whole age range.

Applying these coefficients to specific values makes the pattern concrete. At a BMI of 25, smoking is associated with roughly a 247% increase in expenditure; at the sample mean BMI of 30.66, roughly 364%; at a BMI of 35, roughly 478%. By age, the estimated smoking effect runs from about 645% at age 25, to 364% at the mean age of 39.22, down to about 174% at age 55 and 103% at age 64.

Table 4.4.1. Estimated Smoking Effect on Expenditure at Selected BMI and Age Values

BMI (age held at mean)	Est. % Increase	Age (BMI held at mean)	Est. % Increase
25.00	247%	25	645%
30.66 (mean)	364%	39.22 (mean)	364%
35.00	478%	55	174%
		64	103%

5. Diagnostic Tests

5.1 Multicollinearity

Variance Inflation Factors (VIF) were checked for all predictors, including the interaction terms. All values were low, ranging from 1.0069 to 1.1802, indicating no meaningful multicollinearity, including among the interaction terms themselves.

5.2 Heteroskedasticity

The Breusch-Pagan test was statistically significant (BP = 69.619, $df = 10$, $p = 5.251e-11$), indicating heteroskedasticity in the residuals.

5.3 Robust Standard Errors

Because heteroskedasticity was detected, HC1 robust standard errors were used for inference on the final model. The robust standard errors were close in magnitude to the conventional OLS standard errors, and every previously significant variable remained significant.

5.4 Ramsey RESET

The Ramsey RESET test was statistically significant ($\text{RESET} = 55.597$, $\text{df1} = 2$, $\text{df2} = 1324$, $p < 2.2\text{e-}16$), pointing to possible functional-form misspecification or omitted nonlinear relationships. This result is reported directly rather than set aside: the linear specification may not fully capture the true relationship between the predictors and expenditure, and future work could test nonlinear terms or alternative functional forms.

Table 5.1. Summary of Diagnostic Tests

Test	Result	Interpretation
VIF (max)	1.18	No meaningful multicollinearity
Breusch-Pagan	$\text{BP} = 69.62$, $p < .001$	Heteroskedasticity detected — robust SEs used
Ramsey RESET	$F = 55.60$, $p < .001$	Possible functional-form misspecification
Robust SE	HC1	Applied throughout Sections 4.2–4.4

6. Robustness Check: Gamma GLM

A Gamma Generalized Linear Model with a log link was estimated, since healthcare expenditure is positive and right skewed.

Variable	OLS	Gamma GLM	Conclusion
Age	0.041534	0.034345	Positive and significant in both
BMI	0.001177	0.002176	Not significant in either
Smoking	1.533809	1.425487	Positive and significant in both
Smoking $\times$ BMI	0.051016	0.049948	Positive and significant in both
Smoking $\times$ Age	−0.033361	−0.026441	Negative and significant in both

The Gamma GLM produces broadly consistent findings. Smoking and age stay positive and significant, BMI stays insignificant on its own, and both interaction terms keep their sign and significance. This supports the robustness of the main findings to an alternative model specification, though it does not remove the limitations of observational data or resolve the significant RESET result.

7. Key Findings

- Smoking is strongly and positively associated with healthcare expenditure.
- Age has a positive, statistically significant association with expenditure (roughly 4.2% per year).
- BMI is not statistically significant as a standalone term in the final model.
- The smoking $\times$ BMI interaction is positive and significant: the smoking association strengthens as BMI rises.

- The smoking $\times$ age interaction is negative and significant: the smoking association weakens as age rises.
- The final model explains 82.46% of the variation in log healthcare expenditure.
- Heteroskedasticity is present; HC1 robust standard errors are used throughout.
- The Gamma GLM produces broadly consistent results.
- The significant RESET test suggests the functional form could be refined in future work.

8. Limitations

- The data are observational and cross-sectional, so the results describe associations, not causal effects.
- The dataset contains a limited set of explanatory variables, so other determinants of healthcare expenditure may not be captured.
- The significant RESET test points to possible functional-form misspecification.
- The dataset is not nationally representative; findings should be read within the context of this sample.

9. Conclusion

This project examined how smoking, BMI, and age relate to healthcare expenditure using an applied econometric framework. The final model shows strong positive associations for smoking and age, while BMI carries no independent significance. The smoking $\times$ BMI and smoking $\times$ age interactions are both significant, which means the estimated smoking association is not fixed: it strengthens with BMI and weakens with age.

Heteroskedasticity was detected and addressed with HC1 robust standard errors, and the Gamma GLM produced consistent results under an alternative specification. The significant RESET test remains a real limitation and a clear next step for model refinement.

The workflow, from data cleaning through exploratory analysis, regression modelling, interaction effects, diagnostics, and robustness checks in R, gives a reasonably complete picture of what drives expenditure in this sample, while leaving open exactly which nonlinear relationships the linear specification is missing.